

National University of Singapore

MA3236—Non-Linear Programming

Midterm Test

Semester 1, 2025/2026

Time allowed: 1 hour 15 min

INSTRUCTIONS TO STUDENTS

1. Write down your **student number** on the right and using ink or pencil, shade the corresponding circle **completely** in the grid for each digit or letter. **DO NOT WRITE YOUR NAME!**
2. The assessment paper contains **FOUR (4) questions** and comprises **NINE (9) pages** including this cover page.
3. The time allowed for solving this test is **1 hour 15 min.**
4. Students are required to answer ALL questions.
5. Students should write the answers for each question on the given spaces.
6. This is a **CLOSED BOOK** assessment.
7. Students are allowed to use one A4-sized helpsheet.
8. Calculators are allowed. However, students should lay out systematically the various steps in the calculations.

STUDENT NUMBER												
A												
U	☐	0	0	0	0	0	0	0	0	A	N	
A	☒	1	1	1	1	1	1	1	1	B	R	
HT	☐	2	2	2	2	2	2	2	2	E	U	
NT	☐	3	3	3	3	3	3	3	3	H	W	
		4	4	4	4	4	4	4	4	J	X	
		5	5	5	5	5	5	5	5	L	Y	
		6	6	6	6	6	6	6	6	M		
		7	7	7	7	7	7	7	7			
		8	8	8	8	8	8	8	8			
		9	9	9	9	9	9	9	9			

For Examiner's Use Only

Qns	Marks
Q1	/ 14
Q2	/ 12
Q3	/ 12
Q4	/ 12
Total	/50

1. [14 points] Consider the following problem (P):

$$(P) \quad \min_{x,y,z \in \mathbb{R}} \quad x - 2y - z^2 \\ \text{s.t.} \quad \begin{aligned} 2x - y &\leq 3, \\ -x + y &\leq 1, \\ x &\geq 0, y \geq 0, -5 \leq z \leq 3. \end{aligned}$$

- Show that (P) can be decomposed into a one-dimensional problem (P1) and a two-dimensional problem (P2).
- Solve (P1) analytically / graphically.
- Draw the feasible region of (P2) and hence find all the optimal solutions of (P2).
- Hence, solve the original problem (P).
- What happens if we change the objective function to “ $3y + z^2$ ”? What is the optimal solution (if any) and the optimal objective value?

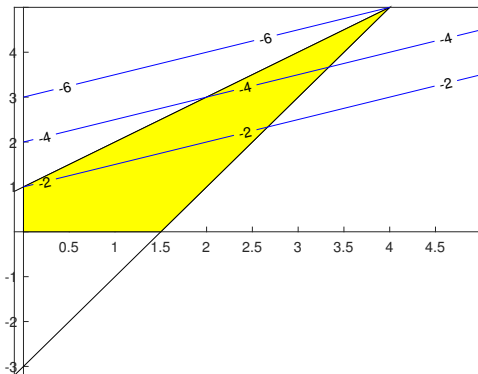
Solution.

- The problem can be separated in terms of (x,y) and z :

$$(P1) \quad \min_{z \in \mathbb{R}} \quad -z^2 \\ \text{s.t.} \quad -5 \leq z \leq 3.$$

$$(P2) \quad \min_{x,y \in \mathbb{R}} \quad x - 2y \\ \text{s.t.} \quad \begin{aligned} 2x - y &\leq 3, \\ -x + y &\leq 1, \\ x &\geq 0, y \geq 0. \end{aligned}$$

- The smallest value of $-z^2$ in the interval $[-5, 3]$ is -25, where the optimal solution is $z^* = -5$.
- The optimal solution of (P2) is (4,5), and the optimal value is -6.



- The optimal solution of the original problem is (4,5,-5), with the optimal value $-6 - 25 = -31$.
- The optimal solution of (P1) becomes 0 since z^2 is nonnegative and the smallest value achieved at 0. For the problem (P2), changing the objective function to $3y$ changes the solution to the set of $\{(x,0) | 0 \leq x \leq 1.5\}$. since y is lower bounded by 0 in the feasible region. Hence, the new optimal solution is the set $\{(x,0,0) | 0 \leq x \leq 1.5\}$ and the optimal value is 0.

2. [12 points] A set $S \subseteq \mathbb{R}^n$ is midpoint convex if for all $\mathbf{x}, \mathbf{y} \in S$, the midpoint between $\mathbf{x}$ and $\mathbf{y}$ is also in S , i.e.

$$\mathbf{x}, \mathbf{y} \in S \Rightarrow \frac{1}{2}(\mathbf{x} + \mathbf{y}) \in S.$$

- (a) Show that the set S defined by

$$S = \{\mathbf{x} \in \mathbb{R}^n \mid \|\mathbf{x}\| \leq 1\}$$

is midpoint convex.

- (b) Show that if S is convex, then it is also midpoint convex.
(c) Conversely, show that if S is a closed midpoint convex set, then it is also convex.
(Hint. You may assume that for an arbitrary $\lambda \in [0, 1]$, there exists a sequence $\lambda_k = c_1 2^{-1} + c_2 2^{-2} + \dots + c_k 2^{-k}$, where $c_i \in \{0, 1\}$, with $\lambda_k \rightarrow \lambda$ as $k \rightarrow \infty$.)

Solution.

- (a) For any $\mathbf{x}, \mathbf{y} \in S$, we have

$$\left\| \frac{1}{2}(\mathbf{x} + \mathbf{y}) \right\| \leq \frac{1}{2}(\|\mathbf{x}\| + \|\mathbf{y}\|) \leq \frac{1}{2}(1 + 1) = 1$$

Hence, $\frac{1}{2}(\mathbf{x} + \mathbf{y}) \in S$ and implying S is midpoint convex.

- (b) Suppose S is convex. Then for any $\mathbf{x}, \mathbf{y} \in S$ and $\lambda \in [0, 1]$, we have

$$\lambda \mathbf{x} + (1 - \lambda) \mathbf{y} \in S.$$

By taking $\lambda = \frac{1}{2}$, we have showed that S is also midpoint convex.

- (c) Suppose S is closed and midpoint convex. Let $\mathbf{x}, \mathbf{y} \in S$ and $\lambda \in [0, 1]$. Choose a sequence

$$\lambda_k = c_1 2^{-1} + c_2 2^{-2} + \dots + c_k 2^{-k}, \quad c_i \in \{0, 1\},$$

such that $\lambda_k \rightarrow \lambda$ as $k \rightarrow \infty$. Then

$$\lambda_k \mathbf{x} + (1 - \lambda_k) \mathbf{y} \rightarrow \lambda \mathbf{x} + (1 - \lambda) \mathbf{y}.$$

Hence, if we can show $\lambda_k \mathbf{x} + (1 - \lambda_k) \mathbf{y} \in S$ for all k , closedness of S implies $\lambda \mathbf{x} + (1 - \lambda) \mathbf{y} \in S$.

Induction step. We prove by induction on k that $\lambda_k \mathbf{x} + (1 - \lambda_k) \mathbf{y} \in S$.

Base case ($k = 1$): Here $\lambda_1 = c_1/2$. If $c_1 = 0$, then $\lambda_1 = 0$ and $\lambda_1 \mathbf{x} + (1 - \lambda_1) \mathbf{y} = \mathbf{y} \in S$. If $c_1 = 1$, then $\lambda_1 = 1/2$ and

$$\lambda_1 \mathbf{x} + (1 - \lambda_1) \mathbf{y} = \frac{1}{2}(\mathbf{x} + \mathbf{y}) \in S$$

by midpoint convexity. So the claim holds for $k = 1$.

Inductive step: Suppose $\mathbf{z}_k := \lambda_k \mathbf{x} + (1 - \lambda_k) \mathbf{y} \in S$. Consider

$$\lambda_{k+1} = \lambda_k + c_{k+1} 2^{-(k+1)}.$$

If $c_{k+1} = 0$, then $\lambda_{k+1} = \lambda_k$ and nothing changes. If $c_{k+1} = 1$, then

$$\lambda_{k+1} \mathbf{x} + (1 - \lambda_{k+1}) \mathbf{y} = \frac{1}{2}(\mathbf{z}_k + (\lambda_k + 2^{-k}) \mathbf{x} + (1 - \lambda_k - 2^{-k}) \mathbf{y}).$$

Note that the second term inside the parentheses corresponds to $\mathbf{z}_k$ with the $(k-1)$ -st digit flipped from 0 to 1, which by the induction hypothesis also lies in S . Therefore, by midpoint convexity, their average belongs to S .

Thus $\lambda_{k+1}\mathbf{x} + (1 - \lambda_{k+1})\mathbf{y} \in S$.

By induction, $\lambda_k\mathbf{x} + (1 - \lambda_k)\mathbf{y} \in S$ for all k . Taking limits and using closedness of S gives

$$\lambda\mathbf{x} + (1 - \lambda)\mathbf{y} \in S.$$

3. [12 points] Consider $f : \mathbb{R}^3 \rightarrow \mathbb{R}$ defined by

$$f(\mathbf{x}) = x_1^3 - 4x_1x_2 + x_2x_3^2 - x_2^2$$

- (a) Find all the stationary points of f in $\mathbb{R}^3$.
- (b) For each stationary point obtained in (a), determine whether it is a local maximizer or a local minimizer or a saddle point of f in $\mathbb{R}^3$.
- (c) Does f have a global minimizer in $\mathbb{R}^3$? Justify your answer.

Solution.

- (a) The stationary points of f can be found by solving the equations:

$$\nabla f(\mathbf{x}) = 0 \Leftrightarrow \begin{pmatrix} 3x_1^2 - 4x_2 \\ -4x_1 + x_3^2 - 2x_2 \\ 2x_2x_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

From the last equation, $x_2 = 0$ or $x_3 = 0$.

(Case i). When $x_2 = 0$, the first equation gives $x_1 = 0$ and hence second equation gives $x_3 = 0$.

(Case ii). When $x_3 = 0$, the second equation gives $x_2 = -2x_1$. Hence, the first equation gives $3x_1^2 + 8x_1 = 0$, implying $x_1 = 0$ or $x_1 = -\frac{8}{3}$.

In summary, there are two stationary points $\mathbf{x}^* = (0, 0, 0)$ and $\mathbf{y}^* = (-8/3, 16/3, 0)$.

- (b) The hessian of f is $H_f(\mathbf{x}) = \begin{pmatrix} 6x_1 & -4 & 0 \\ -4 & -2 & 2x_3 \\ 0 & 2x_3 & 2x_2 \end{pmatrix}$.

For the first stationary point $\mathbf{x}^*$, we have $H_f(\mathbf{x}^*) = \begin{pmatrix} 0 & -4 & 0 \\ -4 & -2 & 0 \\ 0 & 0 & 0 \end{pmatrix}$. This matrix is

indefinite, since the upper block matrix $\begin{pmatrix} 0 & -4 \\ -4 & -2 \end{pmatrix}$ is indefinite because of having negative determinant. So overall, the 3x3 matrix $H_f(\mathbf{x}^*)$ has at least 1 positive and 1 negative eigenvalue. $\mathbf{x}^*$ is a saddle point.

For the second stationary point $\mathbf{y}^*$, we have $H_f(\mathbf{y}^*) = \begin{pmatrix} -16 & -4 & 0 \\ -4 & -2 & 0 \\ 0 & 0 & 32/3 \end{pmatrix}$. This

matrix is indefinite, since the upper block matrix $\begin{pmatrix} -16 & -4 \\ -4 & -2 \end{pmatrix}$ is negative definite by principal minor test. So overall, the 3x3 matrix $H_f(\mathbf{y}^*)$ has at least 1 positive and 1 negative eigenvalue. $\mathbf{y}^*$ is also a saddle point.

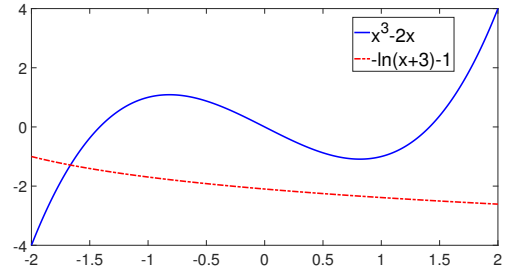
(c) No, there is no global minimizer since there is no local minimizer.

4. [12 points]

Consider the function $f : \mathbb{R} \rightarrow \mathbb{R}$ defined by

$$f(x) = \max\{x^3 - 2x, -\ln(x+3) - 1\}$$

In the following question, you may use the graph of $x^3 - 2x$ and $-\ln(x+3) - 1$ provided here:



Apply golden section method to solve $\min_{x \in \mathbb{R}} f(x)$ by following these steps:

- Find an initial interval $[a_0, b_0]$ with length 1 for which f is unimodal and contains the global minimizer of f .
- Then, manually run one iteration of Golden section search to obtain $[a_1, b_1]$.
- How many iterations n are required to guarantee $|b_n - a_n| < 10^{-5}$?

Solution.

- One appropriate initial interval is $[-2, -1]$. From the graph, we see that f is unimodal in this interval.
- We have $a_0 = -2$, $b_0 = -1$. $\lambda_0 = b_0 - \alpha(b_0 - a_0) = -\frac{\sqrt{5}+1}{2} = -1.618$, $\mu_0 = a_0 + \alpha(b_0 - a_0) = \frac{\sqrt{5}-5}{2} = -1.382$. Comparing the function value, we have $f(\lambda_0) = \max\{-1, -1.3235\} = -1$, $f(\mu_0) = \max\{0.1246, -1.4812\} = 0.1246$. Hence, $f(\mu_0) > f(\lambda_0)$ implying the next interval is $[a_1, b_1] = [a_0, \mu_0] = [-2, -1.382]$.
- The initial interval length is 1. The golden section search shrink the interval by the ratio $\alpha = \frac{\sqrt{5}-1}{2}$ at each iteration, hence after n iterations we have

$$|b_n - a_n| = \alpha |b_{n-1} - a_{n-1}| = \dots = \alpha^n |b_0 - a_0| = \alpha^n.$$

To satisfy the accuracy condition, we require

$$\alpha^n < 10^{-5}.$$

Equivalently,

$$n > \frac{-5}{\log_{10} \alpha} = 23.9.$$

Thus the minimal number of iterations is 24.

End of paper